On the Expressivity of Transformer Encoders

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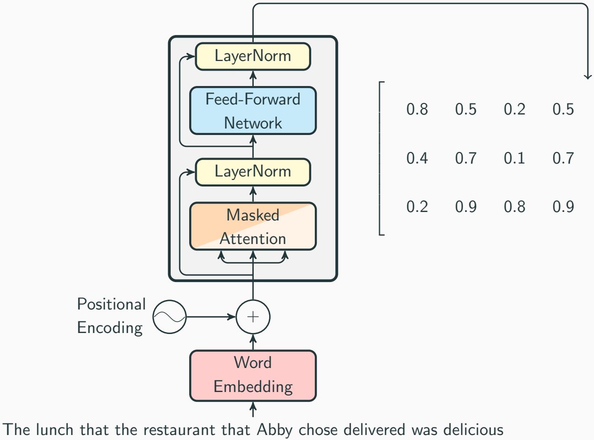
Motivation

"You can't cram the meaning of a whole %&!$# sentence into a single $&!#* vector!" - Ray Mooney.

Sentence of the Day

The lunch that the restaurant that Abby chose delivered was delicious

Transformer Encoders



$$\begin{bmatrix} 0.8 & 0.5 & 0.2 & 0.5 & 0.2 & 0.1 & 0.6 & 0.2 \\ 0.4 & 0.7 & 0.1 & 0.7 & 0.5 & 0.8 & 0.3 & 0.9 \\ 0.2 & 0.9 & 0.8 & 0.9 & 0.7 & 0.4 & 0.7 & 0.8 \end{bmatrix}$$

Sentences and Vectors

$$
\begin{array}{cccccccc}
\left[\begin{array}{cccccccc}
0.8 & 0.5 & 0.2 & 0.5 & 0.2 & 0.1 & 0.6 & 0.2 \\
0.4 & 0.7 & 0.1 & 0.7 & 0.5 & 0.8 & 0.3 & 0.9 \\
0.2 & 0.9 & 0.8 & 0.9 & 0.7 & 0.4 & 0.7 & 0.8
\end{array}\right] \\
\begin{array}{cccccccc}
\uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\
\text{The lunch} & \text{that} & \text{the restaurant} & \text{that} & \text{Abby} & \text{chose} & \text{delivered} & \text{was delicious}
\end{array}
\end{array}
$$

Expressivity

"This sentence involves Abby"

$$\begin{bmatrix} 0.8 & 0.5 & 0.2 & 0.5 & 0.2 & 0.1 & 0.6 & 0.2 \\ 0.4 & 0.7 & 0.1 & 0.7 & \boxed{0.5} & 0.8 & 0.3 & 0.9 \\ 0.2 & 0.9 & 0.8 & 0.9 & \boxed{0.7} & 0.4 & 0.7 & 0.8 \end{bmatrix}$$

Expressivity

"This sentence has a positive tone"

$$\begin{bmatrix} \boxed{0.8 & 0.5 & 0.2 & 0.5 & 0.2 & 0.1 & 0.6 & 0.2} \\ 0.4 & 0.7 & 0.1 & 0.7 & 0.5 & 0.8 & 0.3 & 0.9 \\ 0.2 & 0.9 & 0.8 & 0.9 & 0.7 & 0.4 & 0.7 & 0.8 \end{bmatrix}$$

Expressivity

"This sentence says the lunch was delicious, the restaurant delivered the lunch, and Abby chose the restaurant"

$$\begin{bmatrix} 0.8 & 0.5 & 0.2 & 0.5 & 0.2 & 0.1 & 0.6 & 0.2 \\ 0.4 & 0.7 & 0.1 & 0.7 & 0.5 & 0.8 & 0.3 & 0.9 \\ 0.2 & 0.9 & 0.8 & 0.9 & 0.7 & 0.4 & 0.7 & \boxed{0.8} \end{bmatrix}$$

Sentences and Vectors

$$
\begin{bmatrix}
0.8 & 0.5 & 0.2 & 0.5 & 0.2 & 0.1 & 0.6 & 0.2 \\
0.4 & 0.7 & 0.1 & 0.7 & 0.5 & 0.8 & 0.3 & 0.9 \\
0.2 & 0.9 & 0.8 & 0.9 & 0.7 & 0.4 & 0.7 & 0.8
\end{bmatrix}
$$

$$
\begin{array}{cccccccc}
\uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\
\text{The lunch} & \text{that} & \text{the restaurant} & \text{that} & \text{Abby} & \text{chose} & \text{delivered} & \text{was delicious}
\end{array}
$$

Sentences and Vectors

$$\begin{array}{c} \begin{bmatrix} 0.8 & 0.5 & 0.2 & 0.5 & 0.2 & 0.1 & 0.6 & 0.2 \\ 0.4 & 0.7 & 0.1 & 0.7 & 0.5 & 0.8 & 0.3 & 0.9 \\ 0.2 & 0.9 & 0.8 & 0.9 & 0.7 & 0.4 & 0.7 & 0.8 \end{bmatrix} \\ \begin{matrix} \text{NP} & & \text{NP} & & \text{NP} & \text{VP} & \text{VP} & \text{VP} \end{matrix} \end{array}$$

Sentences and Vectors

$$\begin{array}{cccccccc} \multicolumn{8}{c}{\begin{bmatrix} 0.8 & 0.5 & 0.2 & 0.5 & 0.2 & 0.1 & 0.6 & 0.2 \\ 0.4 & 0.7 & 0.1 & 0.7 & 0.5 & 0.8 & 0.3 & 0.9 \\ 0.2 & 0.9 & 0.8 & 0.9 & 0.7 & 0.4 & 0.7 & 0.8 \end{bmatrix}} \\ {\color{green}(} & & {\color{red}(} & & {\color{blue}(} & {\color{blue})} & {\color{red})} & {\color{green})} \end{array}$$

Dyck Language

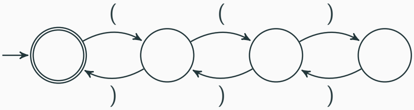
Dyck-1 of depth 3
= strings of parentheses, balanced and nested up to 3 deep
= strings where the number of ('s is equal to the number of)'s, and every prefix contains 0–3 more ('s than)'s

Examples:

- accepted: ((())) ✓
- accepted: (()()) ✓
- rejected: (((()))) ✗
- rejected: ()()(✗

Dyck Language Recognition

Counter-Free Automata



$\textbf{FO}[<]$

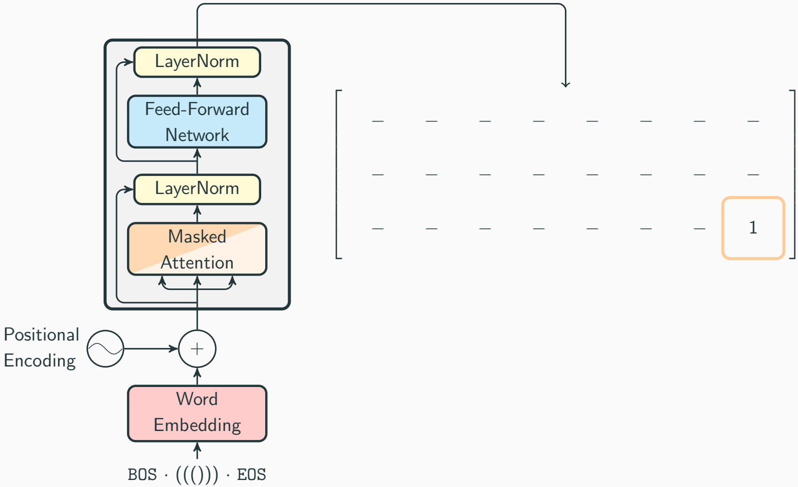
$$\forall x. Q_{(}(x) \rightarrow \exists y > x. Q_{)}(y)$$

$$\vdots$$

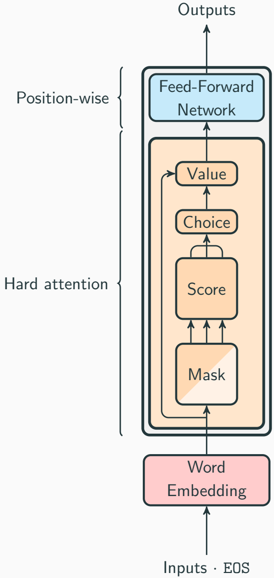
Transformers and Formal Logic

$$\text{Logic} \xrightarrow{\subseteq} \text{Transformers} \xrightarrow{\subseteq} \text{Logic}$$

Transformer Encoders as Formal Language Recognizers



Masked Hard-Attention Transformers



3-layer feed-forward network with ReLU activation

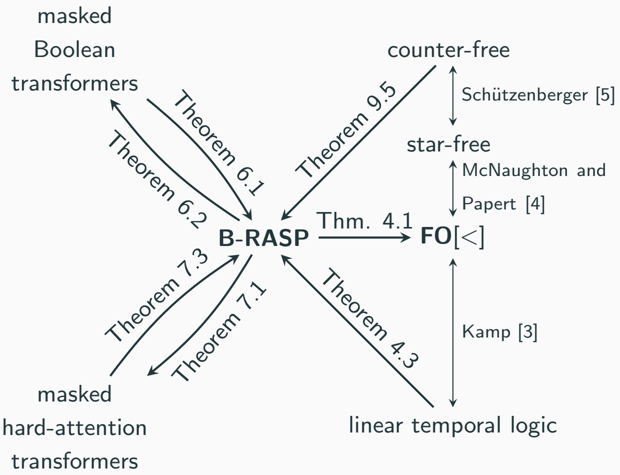
Values: any linear function with residual connection

Leftmost or rightmost maximum

Attention scores: any bilinear function

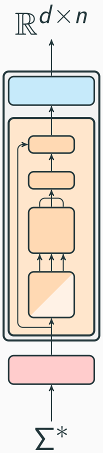
Attention masking: strict future, strict past, or none

B-RASP Results



Transformer Equivalence

masked hard-attention Transformer



B-RASP

$$
\begin{aligned}
& Q_{\texttt{EOS}} \\
& Q_(\\
& Q_) \\
P_((i) &= \blacktriangleright [j < i, 1] \;\; Q_((j) \\
S_)(i) &= \blacktriangleleft [j > i, 1] \;\; Q_)(j) \\
I(i) &= (Q_((i) \wedge S_)(i)) \vee (Q_)(i) \wedge P_((i)) \\
V_((i) &= \blacktriangleleft [j > i, \neg I(j)] \;\; (Q_((i) \wedge \neg I(i) \wedge \neg Q_)(j)) \\
V_)(i) &= \blacktriangleright [j < i, \neg I(j)] \;\; (Q_)(i) \wedge \neg I(i) \wedge \neg Q_((j)) \\
Y(i) &= \blacktriangleleft \left[1, V_((j) \vee V_)(j)\right] \;\; \neg(V_((j) \vee V_)(j))
\end{aligned}
$$

Star-Free Languages

$$\text{Dyck-1 of Depth 3} = \{(), (()), ((())), ()(), (())(), \ldots\}$$

$$(ab)^* = \{\epsilon, ab, abab, ababab, \ldots\}$$

$$\overline{\Sigma^* aa \Sigma^*} = \{\epsilon, a, ab, ba, bb, abb, bab, bba, bbb, \ldots\} = \Sigma^* \setminus \{aa\}$$

Beyond Star-Free?

The lunch that the restaurant that the organizer that NL+ assigned chose delivered was delicious

The lunch that the restaurant that the organizer that the seminar that the CSE department funds assigned chose delivered was delicious

$\vdots$

?

$\mathbf{FOC}^2[+,<;\mathbf{MOD}]$

$$\phi_1 := \exists x. \exists p. (\exists^{=x}_{\leq p} q. Q_{(}(q) \wedge \exists^{=x}_{\leq p} q. Q_{)}(q))$$

"there are the same number of (as)"

$$\phi_2 := \forall p. \exists y. \exists z. (\exists^{=y}_{\leq p} q. Q_{(}(q) \wedge \exists^{=z}_{\leq p} q. Q_{)}(q) \wedge y \geq z)$$

"every prefix has no more) than ("

Then $\phi_1 \wedge \phi_2$ recognizes Dyck-1.

C-RASP

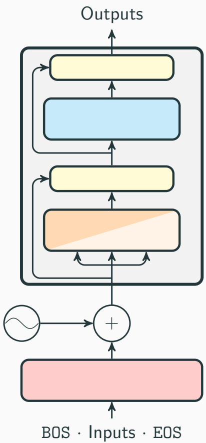
"there are the same number of (as)"

"every prefix has no more) than ("

$$
\begin{aligned}
C_{(}(i) &:= \mathbf{COUNT}_j\,[j < i]\;\; Q_{(}(j) \\
C_{)}(i) &:= \mathbf{COUNT}_j\,[j < i]\;\; Q_{)}(j) \\
V(i) &:= \mathbf{COUNT}_j\,[j \leq i]\;\; C_{(}(i) < C_{)}(i) \\
D(i) &:= (C_{(}(i) = C_{)}(i)) \wedge (V(i) = 0)
\end{aligned}
$$

Transformer Equivalence

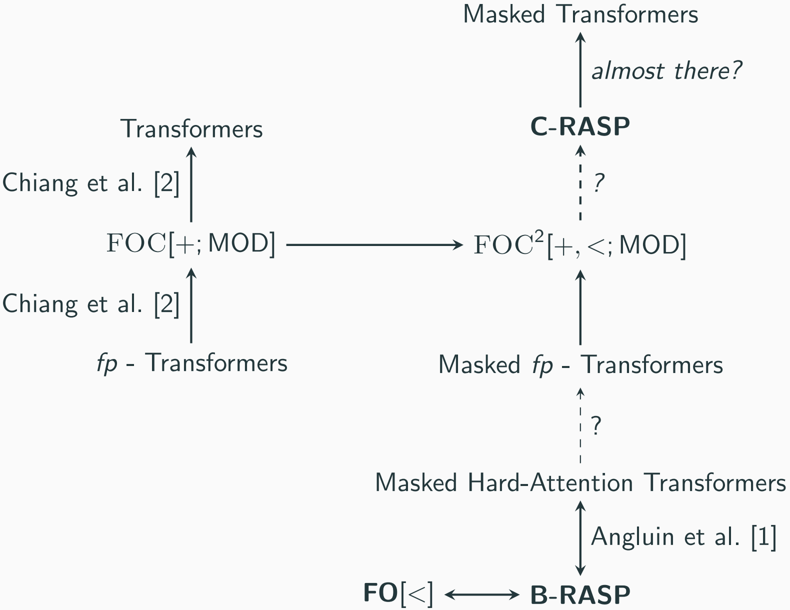
Masked Transformer Encoder



C-RASP

$$
\begin{aligned}
&Q_{\texttt{EOS}} \\
&Q_{(} \\
&Q_{)} \\
C_{(}(i) &= \textbf{COUNT}_j\,[j < i]\;\; Q_{(}(j) \\
C_{)}(i) &= \textbf{COUNT}_j\,[j < i]\;\; Q_{)}(j) \\
V(i) &= \textbf{COUNT}_j\,[j \leq i]\;\; C_{(}(i) < C_{)}(i) \\
D(i) &= (C_{(}(i) = C_{)}(i)) \wedge (V(i) = 0)
\end{aligned}
$$

C-RASP


Masked Transformers
almost there?
C-RASP
?
Transformers
Chiang et al. [2]
FOC[+; MOD]
$\mathrm{FOC}^2[+,<;\mathrm{MOD}]$
Chiang et al. [2]
fp - Transformers
Masked fp - Transformers
?
Masked Hard-Attention Transformers
Angluin et al. [1]
FO[<]
B-RASP

Thank You

Dana Angluin, Stephen Bothwell, David Chiang, Darcey Riley, Ken Sible,

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References

[1] Dana Angluin, David Chiang, and Andy Yang. Masked hard-attention transformers and boolean rasp recognize exactly the star-free languages, 2023.

[2] David Chiang, Peter Cholak, and Anand Pillay. Tighter bounds on the expressivity of transformer encoders. In *Proc. ICML*, 2023.

[3] Johan Anthony Willem Kamp. *Tense Logic and the Theory of Linear Order*. PhD thesis, University of California, Los Angeles, 1968. URL https://www.proquest.com/docview/302320357.

References ii

[4] Robert McNaughton and Seymour Papert. *Counter-Free Automata*. Number 65 in M.I.T. Press Research Monographs. The M.I.T. Press, 1971. ISBN 9780262130769. URL https://archive.org/embed/CounterFre_00_McNa.

[5] M.P. Schützenberger. On finite monoids having only trivial subgroups. *Information and Control*, 8(2):190–194, 1965. DOI 10.1016/S0019-9958(65)90108-7.